Mahima Agarwal

Agarwalmahima01@gmail.com || +91 7599397527

## Education:

10th CBSE | S.K. Public School | 2019| (95.8%)

12th CBSE | S.K. Public School | 2022| (98.8%)

B.Tech(IT &MI) | Cluster Innovation Centre | 2022-26 | Sem 1 : 9.3 CGPA

## Projects:

### *Encryption Decryption :*

This is a simple encryption-decryption project that takes message and applies six different ciphers in a random order to encrypt the message.

The six ciphers used in this project are:

Hill Cipher

Vigenere Cipher

Affine Cipher

Permutation Cipher

Shift Cipher

Substitution Cipher

Github Repo : https://github.com/mahi01agarwal/EncryptionDecryption

### *PPML Using Paillier Cryptosystem:*

This is a Privacy-Preserving Machine Learning (PPML) project that utilizes the Paillier cryptosystem and linear regression to predict laptop prices. With the increasing

concern for privacy in data-driven applications, the proposed approach aims to protect sensitive information while delivering accurate predictions.

The results of the project demonstrate the effectiveness of the PPML approach in predicting laptop prices while preserving privacy. The encrypted predictions can be decrypted to obtain the final predicted prices, providing useful insights to stakeholders without compromising the privacy of the underlying data.

Github Repo : https://github.com/mahi01agarwal/PPML-Using-Paillier

## *Exploring SVD and PCA for Size Reduction in Images and Datasets:*

This project investigates the application of Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) for image and dataset compression. The objective is to reduce the data size while preserving important information. SVD and PCA are implemented using Python libraries, and experiments are conducted to evaluate their effectiveness. The compressed representations achieve significant reduction in storage requirements while retaining key features.

Github Repo : https://github.com/mahi01agarwal/ExploringPCA
https://github.com/mahi01agarwal/ExploringSVD

## *Skills:*

I have basic understanding of Python , Java , C, C++ , HTML , CSS , JS , but the language that I am most proficient is Python.

I know basics of AI/ ML.

I have interest in AI / ML.

## *Contact Me:*

https://www.linkedin.com/in/agarwalmahima01/
agarwalmahima01@gmail.com